2 The quadratic function

What students need to learn	Notes
A The manipulation of quadratic expressions	Students should be able to factorise quadratic expressions and complete the square.
B The roots of a quadratic equation	Students should be able to use the discriminant to identify whether the roots are equal real, unequal real or not real.
C Simple examples involving functions of the roots of a quadratic equation	Students are expected to understand and use: $ax^2 + bx + c = 0$ has roots $\alpha, \beta = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ and forming an equation with given roots, which are expressed in terms of α and β: $\alpha + \beta = \frac{-b}{a}$ and $\alpha\beta = \frac{c}{a}$